

Can Artificial Neural Networks Create Music?



One of the recent applications of neural networks has been in the field of art and music. Are artificial neural networks capable of creativity? Well, they are, although not on par with the human faculty. What are the challenges of creating music and art through algorithms? What are the various approaches? These are some of the questions that are addressed in this article.

Over decades, artificial neural networks (ANNs) have come a long way from single layer *perceptrons* that solved simple classification problems to highly complex networks for deep learning. Today, neural networks are used for more than just classification or optimisation problems in various fields of application. One of the

most recent applications of artificial neural networks is in the field of art and music. The goal here is to simulate human creativity and serve as an assisting tool for artists in their creations.

Early works

One of the first works of using ANNs for music composition was carried out as early as 1988 by Lewis and

Todd. Lewis used a multi-layer *perceptron*, while Todd experimented with a Jordan autoregressive neural network (AR-NN) to generate music sequentially. The music composed with AR-NNs algorithmically had global coherence and structure issues. To circumvent that, Eck and Schmidhuber used long short-term memory (LSTM) neural networks to capture the temporal dependencies in